



PROFESSIONAL CERTIFICATION

CATIA 3D Modelling and Design Training

Master comprehensive 3D modeling and mechanical design in CATIA with part design assembly workbench surfacing drafting and advanced parametric techniques. Create precise complex mechanical components assemblies and production drawings for engineering projects.



35-45

Hours



Certified

Trainers



Industry

Expert



Professional Training | Industry-Recognized Certificate

Dubai, UAE

About Orbit Training Center

Your Gateway to Professional Excellence in Dubai



Who We Are

Established in 2023, Orbit Training Center is a leading professional training provider in Dubai, offering 60+ professional certification courses across diverse fields, including engineering, IT, design, and languages.



Our Benefits

-  Industry recognized certifications
-  Convenient location in the heart of Dubai
-  Affordable fees with multiple payment plans
-  Lifetime post-training support

★Our Features

-  Hands-on learning with real-world projects
-  Expert trainers with industry experience
-  Flexible learning options to fit your schedule
-  State-of-the-art infrastructure & facilities



Our Mission

We deliver outstanding courses, expert teaching, strong career guidance, and placement support to help every student achieve their best results. We stay connected with alumni through networking events, lifelong learning, and ongoing mentorship. We run our operations with smart systems and data-driven decisions to create a smooth, exceptional experience for all.

CATIA 3D Modelling and Design Course

Learn full CATIA workflow for professional 3D mechanical modeling assembly and detailed engineering design.

Course Description

This course delivers complete CATIA training focused on 3D part design solid and surface modeling assemblies constraints drafting GD&T kinematics and sheet metal to produce industry-standard mechanical designs ready for manufacturing.

Duration

35-45

Hours of Training

Format

Classroom / Online

Flexible Option

Key Outcomes

- ✓ Create parametric 3D solid parts
- ✓ Build complex mechanical assemblies
- ✓ Apply advanced surfacing techniques
- ✓ Generate detailed production drawings
- ✓ Implement GD&T annotations accurately
- ✓ Perform kinematic simulations
- ✓ Design sheet metal components

Target Audience

- Mechanical engineers starting with CATIA
- Product designers in manufacturing
- Industrial CAD professionals

Certification

- Certificate of Course Completion
- Industry Recognized Certification
- Trained by Certified Professionals

Prerequisites

- Basic understanding of mechanical concepts
- Familiarity with any 3D CAD software
- Awareness of engineering drawings

CATIA 3D Modelling and Design Course

Start building professional mechanical designs with CATIA's powerful 3D modeling tools.

Intermediate

- Covers all Intermediate Levels
- Precise Geometry Creation
- Basic Parameters & Standards
- Understanding of Tools & Uses
- Intermediate Project Standard
- 1 Project at Completion

Duration
35 Hrs

Professional

- Covers all Professional Modules
- 16 Standard Professional Modules
- Learn with Industry Standard
- Professional Tools and Uses
- Multiple Professional Projects
- 3 Projects at Completion

Duration
40 Hrs

Advanced (Customized)

- Covers All Professional Modules
- 16+ Advance Professional Module
- Co-Ordination and Adv. Parameters
- Shop Drawings & As Built Drawings
- Govt. Standard and Submission
- 4 Projects in Different Categories

Duration
45 Hrs

CATIA 3D Modelling and Design Course

Start building professional mechanical designs with CATIA's powerful 3D modeling tools.

01 Navigating CATIA Interface and Workbenches

- > Explore toolbars and specification tree
- > Customize settings and shortcuts
- > Manage documents and sessions
- > Use compass and view controls
- > Access help and CATSettings

02 Creating Precise 2D Sketches

- > Draw lines arcs conics splines
- > Apply geometric and dimensional constraints
- > Use construction geometry effectively
- > Resolve sketch constraint issues
- > Project 3D elements into sketches

03 Building Parametric Solid Features

- > Create pads pockets shafts ribs
- > Apply fillets chamfers drafts
- > Insert holes and patterns
- > Define work planes and axes
- > Edit feature dependencies

04 Mastering Hybrid Modeling Techniques

- > Combine solids and surfaces
- > Use Boolean operations intelligently
- > Manage multi-body parts
- > Organize geometrical sets
- > Heal and repair geometry

05 Developing Advanced Parametric Strategies

- > Define equations and relations
- > Link parameters contextually
- > Implement skeleton-based design
- > Create configurable features
- > Enforce design rules

06 Advanced Surfacing for Quality

- > Generate multi-section surfaces
- > Create swept and blend surfaces
- > Achieve G2/G3 continuity
- > Analyze curvature and reflections
- > Refine surface topology

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07 Free-Form Shape Design Workflows

- > Sculpt organic free-form shapes
- > Create developed surfaces
- > Apply thickness to surfaces
- > Use intersection and projection
- > Optimize surface flow

08 Advanced Sheet Metal Design

- > Create complex stamped parts
- > Define advanced flanges hems
- > Apply corner reliefs louvers
- > Generate parameterized flat patterns
- > Simulate forming features

09 Knowledge-Based Design Automation

- > Build rule-based features
- > Create powercopies and UDFs
- > Use design tables
- > Define checks and reactions
- > Capture corporate standards

10 Managing Complex Assemblies

- > Structure large product trees
- > Apply advanced constraints
- > Use lightweight mode efficiently
- > Perform clash and section analysis
- > Optimize assembly performance

11 Contextual and Top-Down Assembly

- > Implement skeleton methodology
- > Use published elements
- > Define inter-part relations
- > Manage design variants
- > Create configuration management

12 Producing Detailed Drawings

- > Generate multi-view drawings
- > Apply GD&T annotations
- > Create hole and weld tables
- > Customize drawing templates
- > Produce BOM and parts lists

CATIA 3D Modelling and Design Course

Start building professional mechanical designs with CATIA's powerful 3D modeling tools.

13 Preparing Models for Machining

- > Define manufacturing programs
- > Set up tools and parameters
- > Create prismatic operations
- > Generate drilling cycles
- > Simulate basic toolpaths

14 Advanced Multi-Axis Machining Prep

- > Program 3-5 axis strategies
- > Control tool axis orientation
- > Apply collision avoidance
- > Optimize high-speed paths
- > Generate NC setup documents

15 Advanced Rendering and Visualization

- > Apply realistic materials and textures to models
- > Configure lighting environments and HDRI setups
- > Create photorealistic renderings for presentations
- > Generate exploded views and animation sequences
- > Export high-quality visuals for client deliverables

16 Project Data Management and Collaboration

- > Implement PLM integration for design data management
- > Configure ENOVIA for team collaboration workflows
- > Manage design versions and revision control
- > Export data in STEP IGES and native formats
- > Archive completed projects with full documentation

Final Projects and Assessment:

- **Assessment:** Project-based evaluation through real-world projects to assess practical skills and application without theory exams.
- **Tailored Curriculum:** Customized to industry standards, covering essential topics for professional development in relevant fields.
- **Hands-on Projects:** Real-world simulations and interactive tasks to build applicable skills and create professional portfolios.
- **Certification Prep:** Targeted review sessions, practice exercises, and strategies for achieving industry certifications.
- **Ongoing Support:** Access to recorded materials, forums, instructor guidance, and career assistance for long-term growth.

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Muppuli Raja



One of great institute for REVIT BIM course. I completed my training with Mr Ahmed. He is one great trainer in this institute. He has very good knowledge for BIM Training. I completed my REVIT BIM training with orbit and it was very nice learning journey for me. And one of best thing is training center is very flexible for classes.



Arun



Best SolidWorks training in Dubai! The instructor made 3D modeling and assembly design easy to understand. The course covered all essential tools I need for product design and engineering work. Best training center for solidworks training. Best Training Center for Best Skills



Harshu CH



My experience at Orbit training center has been the best In every step I learn from there very clearly almost complete my course it's very interesting the trainer is the best professional and friendly ,and the rest of the team keep it up, thanks for supporting me Orbit and all staff...



Ranjan Axalan



I just finished my course in 3ds max. I learned a lot on how to create 3d model. Thank you for the experience and for the teaching method it help me to understand easily and grow my knowledge. I will use my learnings to my future company. Highly Recomanded!



Numan



I had good support from trainer Mr Ahmed and the other staff. They would schedule my BIM classes as per my convenience and would retake classes if I required. They always answer my calls when I need assistance. Not only that the trainer would go a mile a ahead by helping me with my work related issues like a true mentor.



Lelanie Fernandez



Overall, I highly recommend the Fusion 360 training at Orbit Center, especially with Ahmad as the instructor. This course has exceeded my expectations, and I am learning a lot. I would rate it 5 out of 5 stars!

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Aman Bakhtiyar



I had a great learning experience at Orbit Training Center. The trainers are knowledgeable, supportive, and always ready to help. The clear and practical teaching methods really helped improve my skills and confidence. I highly recommend Orbit Training Center to anyone looking to learn AutoCAD, Revit, and other professional skills.



Sana Zahid -



I recently completed the online Digital Marketing course with Orbit Training Center, and it was a great experience! The lessons were clear, practical, and easy to follow. The trainers were knowledgeable and supportive throughout the course. I've learned a lot that I can apply to my work here in Belgium. Highly recommend Orbit Training Center to anyone looking to improve their digital marketing skills!



Cleavon Henderson



Orbit Training Center offers outstanding CAD classes that are easy to understand, even as a first-year mechanical engineering student. The instructors simplify complex concepts and use practical examples, making the learning process engaging and effective. Whether you're new to CAD or improving your skills, Orbit Training Center provides a supportive and enriching environment. Highly recommended!



Danish Khan



Just finished Primavera P6 training at Orbit Training Center and it was fantastic! The trainer made complex scheduling concepts easy to understand with real-world examples. Got hands-on experience with project planning, resource allocation, and progress tracking. The course was well-structured and I feel confident using Primavera P6 for my projects now. Great value and would definitely recommend to other project managers!



500+ Students Trained

Join our growing community of professionals

95%

Satisfaction Rate

60+

Courses Offered

4.6

Average Rating



GET STARTED TODAY

Get in Touch with Us

At Orbit Training Center, we offer a diverse range of professional courses designed to equip you with the latest industry skills and knowledge. Our comprehensive programs cover in-demand fields such as Interior Design, Computer Science, Graphic Design, Digital Marketing, Solar Energy, and specialized software training including STAAD Pro, Revit Architecture, Tekla Structures, and more. Each course is crafted by industry experts to provide hands-on learning, real-world applications, and certification preparation.

Up to 30%

Early Bird Discount

Free

Demo Session



Call Us Now

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Contact Information



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